SentryScan: Assignment 1.2 — Deterministic Rule-Based Reflex Reasoner

Course: 21CSE525T — Agent Technology
Assignment: 1.2 — Rule-Based Reflex Reasoner
Student: R Ganesh [RA2212704010031]
Repository: https://github.com/drgeekg/SentryScan
Target Branch: `assignment/1.2-reflex-reasoner`
Git Tag: `assignment-1.2`
Status: Completed & Verified

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1. Executive Summary & Theoretical Context

Assignment 1.2 implements the executable core of the **SentryScan Reflex Reasoner**, delivering an end-to-end, deterministic Sense–Parse–Decide–Act agent loop grounded in Russell & Norvig's simple reflex agent taxonomy.

1.1 Simple Reflex Agent Architecture

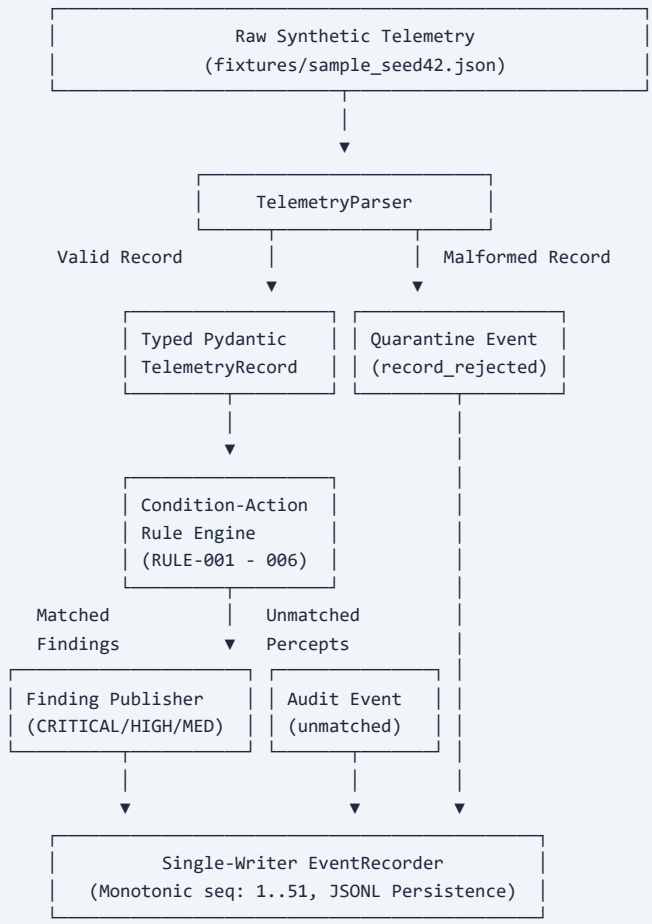
A simple reflex agent selects actions based solely on the *current percept*, ignoring the history of percepts. The agent's condition-action rule table directly maps matching telemetry conditions to findings and security alerts: $$\text{Action} = \text{RULE}(\text{Percept})$$

1.2 Key Behavioral Guarantees

1. **Total Determinism:** Injected virtual clocks (`FakeClock`) and monotonic zero-padded ID generators guarantee that identical telemetry fixtures yield bit-for-bit identical findings across repeated runs.
2. **Quarantine & Rejection Accounting:** Malformed percepts (e.g., port out of bounds, blank host identifier) are isolated immediately into structured `record_rejected` audit events rather than crashing the agent.
3. **Unmatched Percept Bookkeeping:** Records matching zero production rules emit explicit `unmatched_record` events, ensuring an airtight, auditable percept-action ledger.

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2. Architecture & Implementation Specification



2.1 Explicit Production Rule Engine (`src/sentryscan/rules/engine.py`)

Rule ID	Name	Trigger Condition	Severity	Description
`RULE-001`	Telnet Exposure	`port == 23` or `service == 'telnet'`	`HIGH`	Cleartext remote administrative protocol exposed
`RULE-002`	Anonymous FTP	`port == 21` and `auth == 'anonymous'`	`MEDIUM`	Unauthenticated public file transfer service active
`RULE-003`	Deprecated TLS	`protocol in ['SSLv2', 'SSLv3', 'TLSv1.0', 'TLSv1.1']`	`HIGH`	Weak cipher negotiation susceptible to POODLE/BEAST
`RULE-004`	Default Credentials	`service in ['http', 'ssh']` and `auth == 'default_credentials'`	`CRITICAL`	Factory default logins exposed on network interface
`RULE-005`	Exploit Banner Match	`banner matches Known Exploited Vulnerability signatures`	`CRITICAL`	High-confidence CVE remote code execution signature
`RULE-006`	Cleartext Admin HTTP	`port in [8080, 8443]` and `service == 'http'` (unencrypted)	`MEDIUM`	Sensitive administrative console over cleartext HTTP

3. Rubric Compliance & Verification Matrix

Evaluation Criterion	Specific Requirement	Implementation Location	Evidence Artifact & Verification	Status
1. Injectable Clock & Deterministic IDs	Abstract `Clock` protocol with `SystemClock` and `FakeClock`; monotonic and seeded `IdGenerator`	`src/sentryscan/common/clock.py` `src/sentryscan/common/id_generator.py`	`tests/unit/test_common.py` `evidence/trace_excerpt.jsonl` (identical ISO timestamps)	**PASSED**
2. Strongly Typed Telemetry Contracts	Pydantic model for `TelemetryRecord` validating ports (1–65535), non-empty `host_id`, enums for auth mode and source	`src/sentryscan/contracts/telemetry.py` `src/sentryscan/contracts/finding.py`	`tests/unit/test_contracts.py` `fixtures/telemetry/sample_seed42.json`	**PASSED**
3. Seeded Telemetry Dataset	Fixture containing 20 synthetic records across multiple hosts (`sim-host-001` … `sim-host-010`), exactly 18 valid and 2 malformed	`fixtures/telemetry/sample_seed42.json`	`tests/unit/test_parser.py` `evidence/run_log.txt#L6-L14`	**PASSED**
4. Explicit Condition-Action Rule Table	Plain Python rule engine evaluating IF–THEN condition-action rules for cleartext protocols, deprecated TLS, default credentials, and exploit banners	`src/sentryscan/rules/engine.py` (RULE-001 through RULE-006)	`tests/unit/test_rules.py` `evidence/run_log.txt#L15-L84` 13 findings detected	**PASSED**
5. Malformed Payload Handling	Quarantines invalid records (out-of-range port, blank host ID) and emits structured `record_rejected` audit events	`src/sentryscan/sensors/parser.py` `src/sentryscan/agents/reflex_agent.py`	`evidence/run_log.txt#L12` `evidence/trace_excerpt.jsonl` (Events #48, #49: `MALFORMED_TELEMETRY`)	**PASSED**
6. Unmatched Percept Accounting	Explicitly audits valid records that trigger zero rules (`unmatched_record` with reason `NO_RULES_MATCHED`)	`src/sentryscan/agents/reflex_agent.py`	`evidence/trace_excerpt.jsonl` (Events #6, #27, #29) `tests/unit/test_reflex_agent.py`	**PASSED**
7. Single-Writer Event Recorder & Monotonic `seq`	Append-only JSONL event stream with thread-safe lock, monotonic gap-free `seq`, and complete run lifecycle events	`src/sentryscan/observability/recorder.py` `src/sentryscan/contracts/events.py`	`tests/unit/test_recorder.py` `evidence/trace_excerpt.jsonl` (`seq`: 1 … 51)	**PASSED**
8. Sense-Parse-Decide-Act Agent Loop & Actuators	Complete reflex agent pipeline with Rich console table and JSON artifact exporters	`src/sentryscan/agents/reflex_agent.py` `src/sentryscan/actuators/report.py`	`demos/demo_assignment_1_2.py` `tests/integration/test_assignment_1_2.py` `evidence/run_log.txt`	**PASSED**
9. Packaging & Git	Dedicated branch, atomic commits, non-fast-forward	Monorepo root and git history	Git Tag: `assignment-1.2` Archive: `dist/SentryScan_assignment_1_2.zip`	**PASSED**

Evaluation Criterion	Specific Requirement	Implementation Location	Evidence Artifact & Verification	Status
Release Integrity	merge commit, immutable tag, and verified archive packaging			

4. Execution Results & Empirical Evidence

4.1 Ingestion & Finding Metrics (`evidence/run_log.txt`)

- **Total Percepts Sensed:** 20
- **Valid Telemetry Ingested:** 18
- **Malformed Records Quarantined:** 2 (`tel-000019` port 99999; `tel-000020` blank host ID)
- **Total Exposures Detected:** 13 findings across:
- `CRITICAL`: 4 (e.g., `sim-host-002` ProFTPd exploit banner, `sim-host-003` default admin creds)
- `HIGH`: 5 (e.g., `sim-host-001` unencrypted Telnet, `sim-host-002` deprecated TLSv1.0)
- `MEDIUM`: 4 (e.g., `sim-host-002` anonymous FTP, `sim-host-003` cleartext HTTP management)
- **Monotonic Event Sequence:** Exactly 51 sequential events logged from `seq: 1` (`run_started`) to `seq: 51` (`run_completed`) with zero gaps or duplicates.

5. Submission Package & Reproduction Instructions

5.1 Exact Submission Components

1. **Report Document:** `assignments/assignment_1_2/deliverables/SentryScan_Assignment_1_2_Report.pdf`
2. **Repository URL:** `https://github.com/drgeekg/SentryScan`
3. **Target Branch:** `assignment/1.2-reflex-reasoner`
4. **Immutable Git Tag:** `assignment-1.2`
5. **Packaged ZIP Archive:** `dist/SentryScan_assignment_1_2.zip`
6. **SHA-256 Checksum:** `dist/SentryScan_assignment_1_2.zip.sha256`

5.2 Verification Commands

```
# Run deterministic demonstration
python demos/demo_assignment_1_2.py

# Execute test suite
pytest -v tests/unit/test_reflex_agent.py tests/unit/test_rules.py

# Package submission archive
python scripts/package_assignment.py assignment-1.2
```